

permissions entities have over the system. In a public BigChainDB, any participant is able to access the network or deploy that participant's own MongoDB+Tendermint node and connect it to the database federation, while a permissioned BigchainDB could be managed by a consortium or a governing entity, where every member of the consortium manages their own node in the network and no one can join without permission.

BigChainDB's transaction model is analogous to that of Bitcoin as shown

in Figure 3, in the sense that an asset transaction receives an asset input, which is then transformed to an output that may be used in the future as an input for a new transaction. Asset outputs can only be used once as input for a transaction. There are two types of transactions in BigChainDB, as given below.

Create transactions generate a new asset in the system (as a JSON document in MongoDB) with two types of information in it. These are asset information,

which is immutable and can't be modified once the asset is created; and metadata, which can be modified through subsequent *transfer* transactions.

Transfer transactions allow the transfer of ownership of an asset or the modification of the metadata. The only one entitled to perform this transaction over an asset is its owner. These transactions use as input an unused output of the asset, generating as a result a new output with the corresponding modifications.

Key highlights

- *Activity:* Active on GitHub
- *Type of ledger:* Multi-ledger integration
- *Pricing:* Open source
- *Supported languages:* JavaScript and Python
- *GitHub repo:* bigchaindb

HydraChain

HydraChain, an extension of the Ethereum platform, is fully compatible with the Ethereum Protocol on an API and contract level. It supports distributed ledgers and private chains, which are mainly set up for the financial industry. The infrastructure of HydraChain allows users to develop smart contracts in Python, which can improve development and debug efficiency enormously.

HydraChain has a well-defined configuration system, as shown in Figure 5, which provides flexible customisation adjustments such as transaction fees, gas limits, genesis allocation or block time.

Features

- Fully compatible with the Ethereum Protocol
- Accountable validators and native contracts
- Fully customisable, easy to deploy, and open source with commercial support available

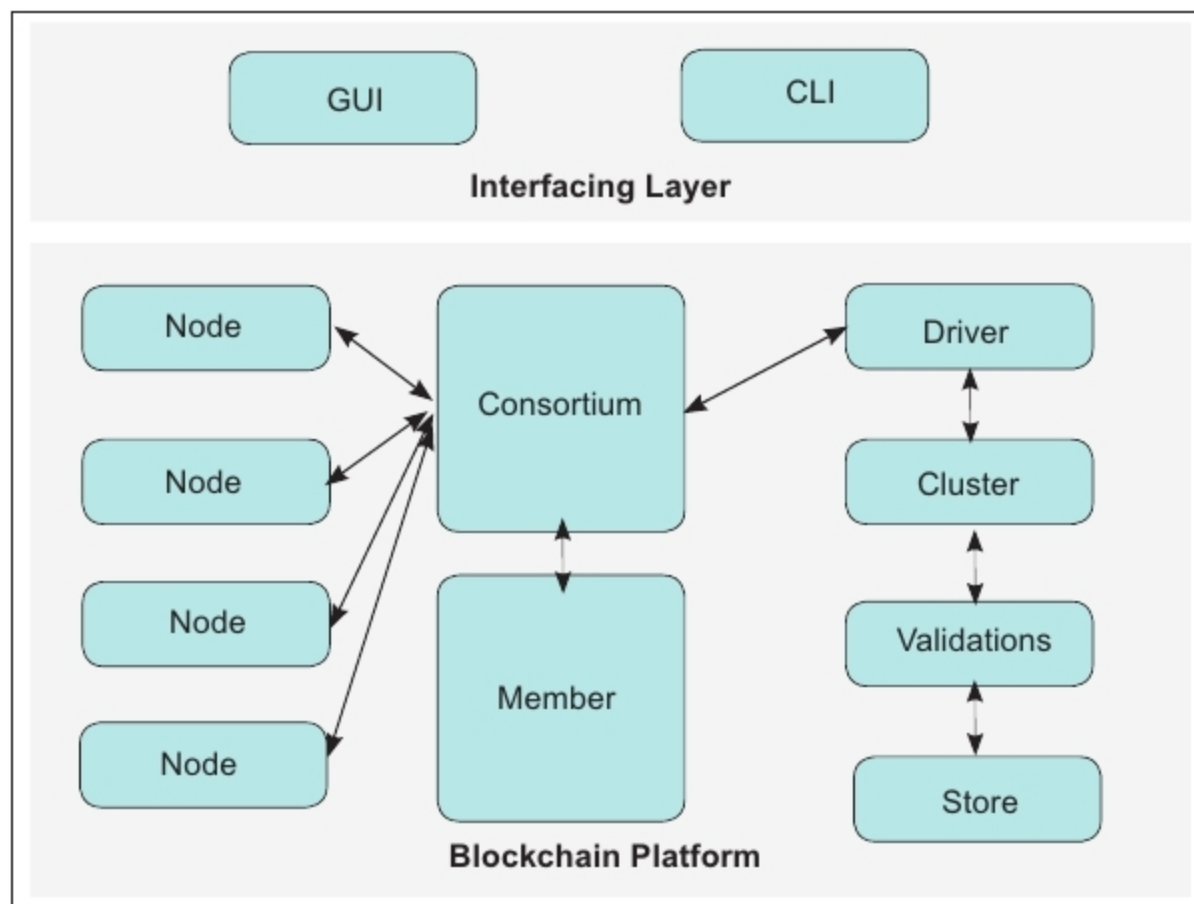


Figure 4: BigChainDB architecture

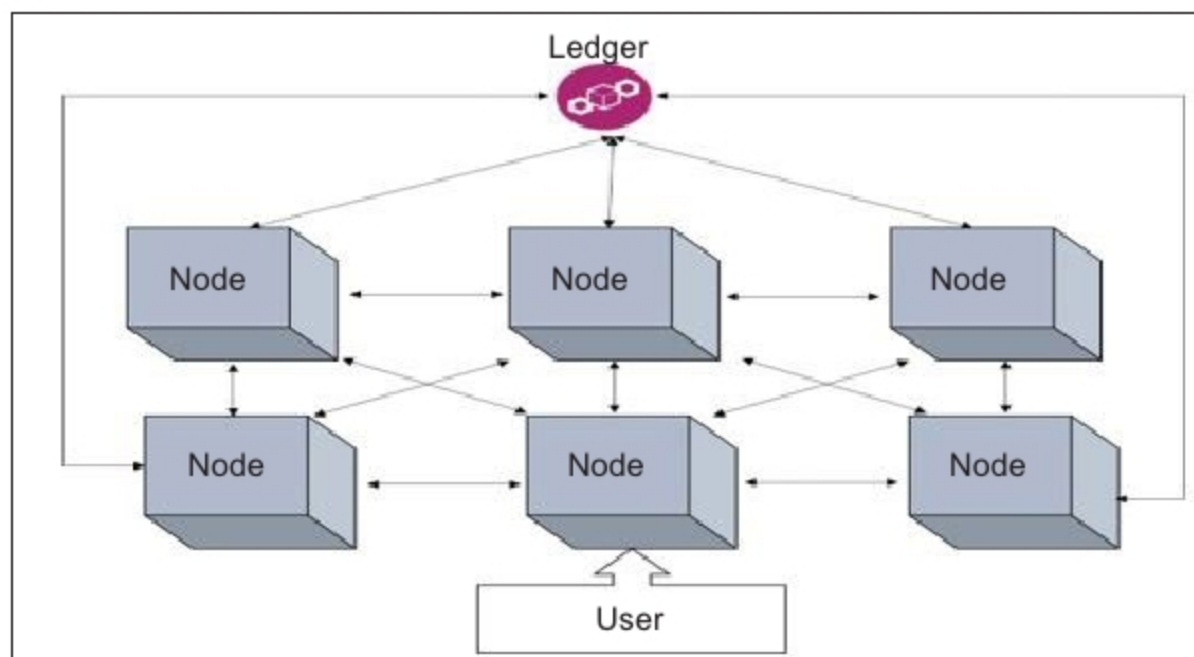


Figure 5: Hydrachain architecture